

45 Days Curriculum

Python for AI

PYTHON FOR AI

Module 1: Python Programming Fundamentals

- Introduction to AI and Python
- Google Colab & Development Environment Setup
- Variables, Data Types, Operators
- Conditional Statements and Loops
- Functions and Modular Programming
- Object-Oriented Programming (OOP)
- Exception Handling and File Handling

Task: Student Result Management

Module 2: Libraries in Python

- NumPy Fundamentals and Advanced Operations
- Pandas Series and DataFrames
- Data Cleaning and Transformation
- Data Aggregation and Merging
- Exploratory Data Analysis (EDA)
- Data Visualization using Matplotlib and Seaborn

Task: Netflix Content Analysis

Module 3: Error knowledge & GUI

- Knowing and understanding different types of Error
- Approaches to resolve error
- GUI Development using Tkinter

Task: Tkinter Based Frontend

Module 4: Industry-Based Project

- Problem Identification and Requirement Analysis
- Dataset Collection and Preparation
- Development
- Deployment using TKinter

Module 5: Github & Cloud Deployment

- Git Fundamentals
- GitHub Collaboration
- Introduction to Model Lifecycle
- Industry Development Workflow

Task: Collaborative Portfolio Repository

Data Analysis

DATA ANALYSIS

Module 1: Python Programming for Data Analytics

- Python Environment Setup
- Variables and Data Types
- Conditional Statements and Loops
- Functions
- OOPS Concepts
- Exception Handling

Task: Business Record Processing System

Module 2: Data Manipulation using Numpy & Pandas

- NumPy Fundamentals
- Pandas Series and DataFrames
- Data Cleaning and Preprocessing
- Data Transformation Techniques
- Grouping and Aggregation
- Merging and Joining Datasets
- Exploratory Data Analysis (EDA)

Task: Enterprise Data Cleaning & Analytics Pipeline

Module 3: Data Visualization

- Visualization using Matplotlib
- Interactive Visualizations using Seaborn
- Dashboard Design Best Practices
- Business KPI Analysis
- Predictive Analytics Fundamentals

Task: Problem based Intelligence Dashboard

Module 4: Data Analysis with Excel

- Data Types and Formatting
- Formulas and Functions
- Logical Functions (IF, IFS, AND, OR)
- Data Validation and Data Cleaning
- Advanced Lookup Functions
- Pivot Tables and Pivot Charts

Task: Executive Sales & Performance Dashboard

Module 5 : SQL for Data Analysis

- SQL Fundamentals
- Data Retrieval and Filtering
- Sorting and Aggregation
- Grouping and Summarization
- Table Joins and Relationships
- Window Functions

Task: Sales & Customer Data Analysis using SQL

Module 6: Industry Practices & Project

- Git Fundamentals
- GitHub Collaboration
- Version Control Best Practices
- Portfolio Development
- Cloud Deployment

Data Science

DATA SCIENCE

Module 1: Python Programming for Data Analytics

- Python Environment Setup
- Variables and Data Types
- Conditional Statements and Loops
- Functions and Scope
- OOP Concepts
- Exception Handling

Task: Student Record management System

Module 2: Data Analysis

- NumPy for Numerical Computing and Statistical Analysis
- Data Handling and Transformation using Pandas
- Data Cleaning and Preprocessing Techniques
- Exploratory Data Analysis (EDA) and Insight Generation
- Insight Generation and analysis

Task: Real-World Data Analysis using Kaggle Datasets

Module 3: Data Extraction

- Data Acquisition from Multiple Sources
- Working with JSON Data
- Web Scraping using Requests and BeautifulSoup
- Extracting and Processing Web-Based Information
- Real-World Data Collection Practices

Task: Automated Web Data Collection and Analysis

Module 4: Data Visualization & Business Intelligence

- Data Visualization Fundamentals and Design Principles
- Creating Charts and Graphs using Matplotlib
- Statistical Visualization using Seaborn
- Interactive Dashboards and Visual Analytics with Plotly
- Business Intelligence Reporting using Power BI

Task: Interactive Business Intelligence Dashboard

Module 5: SQL for Data Analysis

- Database Fundamentals and Relational Concepts
- Data Retrieval using SQL Queries

- Data Filtering, Sorting, and Aggregation Techniques
- Working with Joins and Multiple Tables
- Subqueries and Data Analysis Functions
- Business Reporting and Data Insights using SQL
- Window functions in SQL

Task: Sales and Customer Database Analysis

Module 6: GitHub & Cloud Deployment

- Git Fundamentals
- GitHub Repositories
- Version Control
- Project Documentation
- Portfolio Building

Task: Deploy a Real Time Project on Github

Machine Learning

MACHINE LEARNING

Module 1: Python for Machine Learning

- NumPy and Pandas
- Data Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Data Visualization

Task: Data Preprocessing & EDA Pipeline

Module 2: Supervised Machine Learning

- Machine Learning Fundamentals
- Regression Algorithms
- Classification Algorithms
- Cost/Loss Functions
- Gradient Descent Optimization
- Model Training & Testing
- Model Selection Techniques

Algorithms Covered

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Task: Customer Churn Prediction

Module 3: Unsupervised Learning & Feature Engineering

- Clustering Techniques
- Dimensionality Reduction
- Feature Selection
- Principal Component Analysis (PCA)
- Anomaly Detection

Algorithms Covered

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN

- PCA

Task: Customer Segmentation & Pattern Analysis

Module 4: Model Evaluation & Optimization

- Performance Evaluation Metrics
- Cross Validation Techniques
- Hyperparameter Tuning
- Overfitting & Underfitting
- Bias-Variance Tradeoff
- Ensemble Learning Fundamentals
- Model Explainability (Feature Importance)

Task: Machine Learning Model Optimization

Module 5: Industry Project

- Problem Identification
- Dataset Collection & Preparation
- End-to-End Model Development
- Model Evaluation & Optimization
- Project Documentation
- Portfolio Development

Industry Project: End-to-End Machine Learning Solution for a Real-World Business Problem.

Deep Learning

DEEP LEARNING

Module 1: Foundations of Deep Learning

- Introduction to Artificial Intelligence, Machine Learning & Deep Learning
- Evolution and Industry Applications of Deep Learning
- AI vs ML vs DL
- Neural Networks: Biological vs Artificial Neurons
- Components and Workflow of Deep Learning Models
- TensorFlow, PyTorch & Keras Frameworks
- Google Colab, Jupyter Notebook & Environment Setup
- GPU vs CPU, CUDA Basics & Industry Best Practices

Module 2: Mathematical Foundations for Deep Learning

- Linear Algebra: Scalars, Vectors, Matrices & Tensors
- Matrix Operations and Mathematical Foundations
- Calculus Concepts for Deep Learning
- Derivatives, Chain Rule & Gradients
- Gradient Descent and Optimization Techniques
- Cost Functions and Loss Functions
- Convex vs Non-Convex Optimization
- Probability Concepts for Deep Learning

Task: Implement Gradient Descent from Scratch

Module 3: Neural Network Fundamentals

- Logistic Regression and Perceptron Models
- Binary and Multi-Class Classification
- Feed Forward Neural Networks
- Forward & Backward Propagation
- Activation Functions (Sigmoid, ReLU, Tanh, Softmax)
- Loss Functions and Model Optimization
- Weight Initialization and Bias Optimization
- Vanishing & Exploding Gradient Problems

Task: Classification of different objects.

Module 4: Deep Neural Networks

- Deep Neural Network Architecture
- Hidden Layers and Matrix Computations
- Hyperparameters and Batch Processing
- Gradient Descent Variants (Batch, Mini-Batch & SGD)
- Optimizers (SGD, Momentum, RMSProp & Adam)
- Learning Rate Scheduling
- LSTM & GRU Architecture
- Regularization, Dropout & Batch Normalization
- Early Stopping and Performance Optimization

Task: Loan Approval Prediction

Module 5: Industry Project & Deployment

- End-to-End Deep Learning Workflow
- Model Evaluation Metrics and Performance Analysis
- Model Saving & TensorFlow/PyTorch Serialization
- Streamlit & Flask-based Model Deployment
- GitHub, Documentation & Version Control
- Resume Building and Interview Preparation
- Industry Best Practices for AI Deployment

Project: Develop and Deploy a Real-World Deep Learning Solution.

6 Months Fellowship

Foundational AI

FOUNDATIONAL AI

Module 1: Programming with Python

Duration: 3 Weeks

Python Fundamentals & Advance Concepts

- Introduction to Python
- Variables & Data Types
- List, Dictionary & Set Comprehensions
- Input & Output
- Operators
- Conditional Statements
- Loops
- Pattern Programming
- Functions
- Recursion
- Modules & Packages
- File Handling
- Exception Handling
- Classes & Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

Module 2: Data Analysis with Python

Duration: 2 Weeks

NumPy

- Arrays
- Indexing
- Operations
- Vectorization

Pandas

- Series & DataFrames
- Data Cleaning
- Missing Values
- Feature Engineering

- GroupBy
- Merge
- Concatenation

Data Visualization

- Matplotlib
- Plotly
- Statistical Visualizations
- Exploratory Data Analysis (EDA)

Module 3: Data Analytics with Excel

Duration: 1 Week

Excel Fundamentals

- Data Formatting
- Tables
- Named Ranges

Data Analysis

- Sorting
- Filtering
- Conditional Formatting
- Data Validation
- Remove Duplicates
- Flash Fill

Excel Functions

- IF & IFS
- SUMIF & COUNTIF
- VLOOKUP
- XLOOKUP
- INDEX & MATCH
- TEXT Functions
- DATE Functions

Reporting

- Pivot Tables
- Pivot Charts
- Interactive Dashboards
- KPI Reporting

Module 4: Business Intelligence with Power BI

Duration: 1-2 Weeks

Power BI Fundamentals

- Power BI Desktop
- Power Query
- Data Import
- Data Cleaning

Data Modeling

- Relationships
- Star Schema
- Snowflake Schema

DAX

- Calculated Columns
- Measures
- CALCULATE
- FILTER
- IF
- SWITCH
- RANKX
- Time Intelligence

Visualization

- Cards
- Tables
- Matrix
- Charts
- Maps
- Slicers
- KPI Dashboards

Module 5: Machine Learning Engineering

Duration: 3 Weeks

Machine Learning Workflow

- ML Lifecycle
- Data Preprocessing
- Feature Engineering
- Feature Scaling

- Train-Test Split

Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- KNN
- Naive Bayes

Unsupervised Learning

- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA
- Apriori Algorithm

Model Evaluation

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Cross Validation
- Hyperparameter Tuning

Module 6: Deep Learning Engineering

Duration: 4 Weeks

Neural Networks

- Perceptron
- ANN
- Forward Propagation
- Backpropagation
- Activation Functions
- Optimizers
- Regularization

TensorFlow & PyTorch

- Tensor Operations

- Model Building
- Training
- Evaluation

CNN

- Image Classification
- Transfer Learning

RNN

- Sequential Data
- NLP Fundamentals & Preprocessing
- LSTM
- GRU

Introduction to Transformers

- Attention Mechanism
- Transformer Architecture
- BERT vs GPT (Conceptual)

Module 7: FastAPI & Backend Development

Duration: 3-4 Weeks

REST API Fundamentals

- HTTP Methods
- Status Codes
- JSON
- REST Architecture

FastAPI

- Routing
- Path Parameters
- Query Parameters
- Request & Response Models
- Pydantic Validation

Database Integration

- PostgreSQL Integration
- SQLAlchemy / SQLModel
- CRUD APIs

Security

- JWT Authentication
- Password Hashing
- Protected Routes

API Development

- File Upload APIs
- Error Handling
- Swagger Documentation

Module 8: SQL & Database Engineering

Duration: 2-3 Weeks

Database Fundamentals

- DBMS vs RDBMS
- Database Design
- ER Diagrams
- Normalization
- Primary & Foreign Keys
- Constraints

SQL Programming

- CRUD Operations
- Operators
- Joins
- Subqueries
- Views
- Common Table Expressions (CTEs)
- Window Functions
- Stored Procedures (Introduction)
- Functions

Transactions & Database Management

- ACID Properties
- COMMIT
- ROLLBACK
- SAVEPOINT
- Basic Concurrency Concepts

Query Optimization

- Indexes
- Clustered vs Non-Clustered Indexes (Concept)
- EXPLAIN
- EXPLAIN ANALYZE
- Execution Plans
- Query Latency
- Performance Tuning
- Query Rewriting Best Practices

SQL for AI

- Feature Extraction
- Data Cleaning
- Aggregations
- Time-Series Queries

Module 9: AI Application Development

Duration: 2 Weeks

Streamlit

- Widgets
- Session State
- Layouts
- Charts
- Forms

AI Frontend Development

- Connecting Streamlit with FastAPI
- File Upload Interface
- Model Inference
- Dashboard Development

Module 10: AI Deployment & DevOps Fundamentals

Duration: 1 Week

GitHub & Cloud Deployment

- Git Fundamentals
- GitHub Repositories
- Version Control

Docker

- Images
- Containers
- Dockerfile
- Docker Compose

DevOps Basics

- CI/CD Concepts
- Version Control Workflow
- Model Versioning

Agentic AI

AGENTIC AI AND INTELLIGENT SYSTEM ENGINEERING

Module 1: Foundations of Generative AI & Large Language Models

Duration: 2 Weeks

Topics

- Evolution of AI
- Generative AI Fundamentals
- Transformer Architecture
- LLM Architecture
- Decoder-only vs Encoder-Decoder Models
- Tokens & Tokenization
- Embeddings
- Context Windows
- Sampling Techniques
- Prompting Basics
- Model Selection
- Open-source vs Proprietary Models
- API-based vs Local Models
- AI Application Architecture

Module 2: Prompt Engineering & AI Reasoning

Duration: 2 Weeks

Topics

- Prompt Engineering Fundamentals
- Zero-shot Prompting
- One-shot Prompting
- Few-shot Prompting
- Chain-of-Thought Prompting
- Self-Consistency
- Tree of Thoughts (Concept)
- ReAct Pattern
- Role Prompting
- Structured Outputs
- Prompt Templates
- Prompt Optimization
- Prompt Testing

Module 3: AI Agent Fundamentals

Duration: 2 Weeks

Topics

- What are AI Agents?
- Agent Architecture
- Agent Lifecycle
- Planning
- Reasoning
- Tool Calling
- Memory
- Reflection
- Agent Communication
- Autonomous Decision Making
- Human-in-the-Loop Systems
- AI Workflows vs AI Agents

Module 4: Retrieval-Augmented Generation (RAG)

Duration: 3 Weeks

Topics

- RAG Fundamentals
- Document Processing
- Chunking Strategies
- Embeddings
- Semantic Search
- Vector Similarity
- Hybrid Search
- Metadata Filtering
- Reranking
- Context Management
- Graph RAG (Introduction)
- Enterprise Search Concepts

Module 5: Vector Databases & Knowledge Systems

Duration: 2 Weeks

Topics

- Vector Databases
- Embedding Models
- Similarity Search
- Indexing
- Collections
- Metadata
- Knowledge Graphs (Introduction)
- Enterprise Knowledge Systems
- Data Ingestion Pipelines

Module 6: Agent Frameworks & Orchestration

Duration: 3 Weeks

Topics

- LangChain
- LangGraph
- LlamaIndex
- DSPy (Introduction)
- OpenAI Agents SDK
- Workflow Design
- Tool Calling
- Memory Management
- Agent Routing
- Agent State Management

Module 7: Multi-Agent Systems Engineering

Duration: 2 Weeks

Topics

- Multi-Agent Architectures
- Agent Collaboration
- Agent Communication
- Task Delegation
- Coordinator Pattern
- Planner Pattern
- Supervisor Pattern
- Reflection Loops
- Conflict Resolution
- Distributed Agents

Module 8: Enterprise AI Integration & Automation

Duration: 2 Weeks

Topics

- Enterprise APIs
- REST Integration
- OAuth Basics
- Workflow Automation
- Webhooks
- CRM Integration
- ERP Integration
- Email Automation
- Calendar Automation
- Database Integration
- AI Business Process Automation

Module 9: AI Application Engineering

Duration: 2 Weeks

Topics

- FastAPI
- Streamlit
- Authentication
- Session Management
- File Upload Systems
- Chat Interfaces
- API Design
- Backend Architecture
- Frontend Integration
- Project Structure

Module 10: Deployment, LLMOps & Production Systems

Duration: 2 Weeks

Topics

- Docker

- Docker Compose
- CI/CD Concepts
- LLMOps Fundamentals
- Model Versioning
- API Scaling
- Monitoring
- Logging
- Observability
- Cost Optimization
- Caching
- Rate Limiting

Generative AI

AI ENGINEERING & GENERATIVE AI

Module 1: Foundations of Generative AI & Large Language Models

Duration: 2 Weeks

Topics

- Evolution of Artificial Intelligence
- Generative AI Fundamentals
- Transformer Architecture
- Encoder vs Decoder Models
- GPT, BERT, T5, LLaMA, Mistral, Gemma
- Tokenization
- Embeddings
- Context Windows
- Attention Mechanism
- Sampling Strategies
- Model Parameters
- Open-source vs Proprietary Models
- AI Application Architecture

Module 2: Prompt Engineering & LLM Application Design

Duration: 2 Weeks

Topics

- Prompt Engineering Fundamentals
- Zero-shot, One-shot & Few-shot Prompting
- Chain-of-Thought Prompting
- Self-Consistency
- ReAct (Concept)
- Structured Outputs
- JSON Responses
- Function Calling
- Prompt Templates
- Prompt Testing & Evaluation
- AI Output Optimization

Module 3: LLM APIs & AI Application Development

Duration: 2 Weeks

Topics

- OpenAI API
- Gemini API
- Claude API
- Hugging Face Inference API
- API Authentication
- Streaming Responses
- Token Usage
- Cost Optimization
- Error Handling
- API Rate Limits
- Model Selection

Module 4: Embeddings, Vector Databases & Semantic Search

Duration: 2 Weeks

Topics

- Embeddings
- Similarity Search
- Dense vs Sparse Retrieval
- Semantic Search
- Hybrid Search
- Metadata Filtering
- Vector Indexing
- Retrieval Strategies
- Reranking (Introduction)

Module 5: Retrieval-Augmented Generation (RAG)

Duration: 3 Weeks

Topics

- RAG Architecture
- Document Loading
- Chunking Strategies
- Embedding Pipelines
- Retrieval Techniques
- Context Management
- Prompt Augmentation
- Query Rewriting
- Multi-document Retrieval
- Graph RAG (Introduction)
- Enterprise Knowledge Systems

Module 6: Multimodal AI Engineering

Duration: 2 Weeks

Topics

- Vision-Language Models
- Image Understanding
- OCR Integration
- Speech-to-Text
- Text-to-Speech
- Audio Processing
- Image Generation
- Image Captioning
- Video AI (Introduction)
- AI Document Processing

Module 7: Fine-Tuning & Model Customization

Duration: 2 Weeks

Topics

- Fine-Tuning Fundamentals
- Transfer Learning for LLMs
- PEFT
- LoRA
- QLoRA
- Dataset Preparation
- Instruction Tuning
- Model Evaluation
- Quantization (Introduction)
- Distillation (Concept)

Module 8: AI Application Engineering

Duration: 2 Weeks

Topics

- FastAPI
- Streamlit
- Gradio
- Authentication
- Session Management
- File Upload Systems
- API Design

- Backend Architecture
- Frontend Integration
- Project Organization

Module 9: AI Deployment & LLMOps

Duration: 2 Weeks

Topics

- Docker
- Docker Compose
- Environment Management
- Model Versioning
- CI/CD Concepts
- Monitoring
- Logging
- API Scaling
- Caching
- Cost Optimization
- Deployment Best Practices

Module 10: Enterprise AI Solutions & Integration

Duration: 2 Weeks

Topics

- Enterprise AI Architecture
- REST APIs
- Database Integration
- Webhooks
- Authentication
- AI Integration Patterns
- CRM & ERP Integration (Concept)
- Workflow Automation (Introduction)
- AI Product Design
- Scalability Considerations

AI Infrastructure, Cloud & LLMOps Engineering

AI INFRASTRUCTURE, CLOUD & LLMOPS ENGINEERING

Module 1: AI Infrastructure Fundamentals

Duration: 2 Weeks

Topics

- AI Infrastructure Overview
- AI System Architecture
- Cloud Computing Fundamentals
- Virtual Machines
- Containers vs Virtual Machines
- CPU vs GPU vs TPU
- AI Hardware Basics
- Networking Fundamentals
- Linux for AI Engineers
- File Systems
- Environment Variables
- Package Management
- SSH & Remote Access

Module 2: Cloud Computing for AI

Duration: 2 Weeks

Topics

- Cloud Service Models (IaaS, PaaS, SaaS)
- Cloud Storage
- Compute Services
- Networking Basics
- Identity & Access Management (IAM)
- Secrets Management
- Object Storage
- Load Balancers
- Auto Scaling Concepts
- Cost Optimization
- Multi-Cloud Concepts

Module 3: Docker & Containerization

Duration: 2 Weeks

Topics

- Docker Fundamentals
- Docker Images
- Docker Containers
- Dockerfile
- Docker Compose
- Volumes
- Networks
- Environment Variables
- Multi-stage Builds
- Container Best Practices
- Image Optimization

Module 4: Kubernetes & Container Orchestration

Duration: 3 Weeks

Topics

- Kubernetes Architecture
- Pods
- Deployments
- ReplicaSets
- Services
- ConfigMaps
- Secrets
- Persistent Volumes
- Ingress
- Horizontal Pod Autoscaling
- Helm Basics
- Rolling Updates

Module 5: MLOps & LLMOps Foundations

Duration: 2 Weeks

Topics

- MLOps Lifecycle
- LLMOps Lifecycle
- Model Registry
- Experiment Tracking
- Model Versioning
- Prompt Versioning
- Dataset Versioning
- Pipeline Automation

- Reproducibility

Module 6: CI/CD for AI Systems

Duration: 2 Weeks

Topics

- GitHub Actions
- CI/CD Pipelines
- Automated Testing
- Build Automation
- Docker Image Deployment
- Infrastructure Automation
- Release Strategies
- Rollback Strategies

Module 7: Production AI Deployment

Duration: 2 Weeks

Topics

- FastAPI Deployment
- Streamlit Deployment
- Model Serving
- API Gateways
- Reverse Proxy
- Load Balancing
- Scaling AI APIs
- High Availability
- Serverless AI (Introduction)

Module 8: Monitoring, Logging & Observability

Duration: 2 Weeks

Topics

- AI Monitoring
- Infrastructure Monitoring
- API Monitoring
- Prompt Monitoring
- Model Drift
- Data Drift
- Log Management

- Metrics
- Alerting
- Dashboards
- Incident Response

Module 9: AI Performance Optimization

Duration: 2 Weeks

Topics

- GPU Utilization
- Batch Processing
- Request Queuing
- Caching
- Redis
- Rate Limiting
- Response Time Optimization
- Cost Optimization
- Model Quantization (Deployment Perspective)
- Inference Optimization

Module 10: Security & Governance for AI Infrastructure

Duration: 2 Weeks

Topics

- AI Infrastructure Security
- Authentication
- Authorization
- JWT
- API Keys
- Secrets Management
- Network Security
- Encryption Basics
- AI Governance
- Compliance Overview
- Backup & Disaster Recovery

Module 11: Enterprise AI Platform Engineering

Duration: 2 Weeks

Topics

- AI Platform Architecture
- Multi-Tenant Systems
- Workflow Scheduling
- Message Queues (Introduction)
- Event-Driven Architecture
- API Management
- AI Platform Design
- Cost Management
- Scalability Planning
- Enterprise Integration

AI security

AI SECURITY & RESPONSIBLE AI ENGINEERING

Module 1: Foundations of AI Security & Responsible AI

Duration: 2 Weeks

Topics

- Introduction to AI Security
- AI Threat Landscape
- AI Attack Surface
- Secure AI Development Lifecycle
- Responsible AI Principles
- AI Ethics
- Fairness
- Transparency
- Explainability
- Accountability
- AI Risk Management
- AI Governance Overview
- Regulatory Landscape (High Level)

Module 2: Machine Learning Security

Duration: 2 Weeks

Topics

- ML Pipeline Security
- Data Poisoning
- Label Poisoning
- Model Poisoning
- Evasion Attacks
- Model Extraction
- Membership Inference
- Data Leakage
- Privacy Risks
- Defensive Strategies

Module 3: LLM Security

Duration: 3 Weeks

Topics

- LLM Threat Model
- Prompt Injection
- Indirect Prompt Injection
- Jailbreak Attacks
- Prompt Leakage
- System Prompt Protection
- Hallucinations
- Sensitive Data Exposure
- Tool Abuse
- Function Calling Risks
- Secure Prompt Design
- Output Validation

Module 4: AI Red Teaming & Adversarial Testing

Duration: 2 Weeks

Topics

- AI Red Team Methodology
- Threat Modeling
- Attack Simulation
- Adversarial Examples
- Model Robustness Testing
- Security Test Cases
- AI Penetration Testing Concepts
- Vulnerability Reporting

Module 5: AI Evaluation & Model Validation

Duration: 2 Weeks

Topics

- AI Evaluation Fundamentals
- Benchmarking
- Accuracy vs Reliability
- Hallucination Evaluation
- Factuality
- Toxicity Detection
- Bias Detection
- Safety Evaluation
- Prompt Evaluation
- Human Evaluation
- Automated Evaluation

Module 6: AI Safety Engineering & Guardrails

Duration: 2 Weeks

Topics

- AI Guardrails
- Input Validation
- Output Filtering
- Policy Enforcement
- Safety Classifiers
- Content Moderation
- Human-in-the-Loop
- Secure Tool Calling
- Secure Agent Design
- Fail-safe Mechanisms

Module 7: Privacy-Preserving AI

Duration: 2 Weeks

Topics

- Data Privacy Fundamentals
- Personally Identifiable Information (PII)
- Data Anonymization
- Data Masking
- Secure Data Handling
- Differential Privacy (Concept)
- Federated Learning (Introduction)
- Encryption Basics
- Secure Storage
- Access Control

Module 8: Secure AI Application Engineering

Duration: 2 Weeks

Topics

- Secure FastAPI Development
- Authentication
- Authorization
- JWT Security
- API Security
- Secrets Management
- Secure Configuration

- Secure Logging
- Rate Limiting
- Input Sanitization

Module 9: AI Infrastructure Security

Duration: 2 Weeks

Topics

- Container Security
- Docker Security
- Kubernetes Security (Introduction)
- Cloud Security Fundamentals
- Identity & Access Management
- Secrets Management
- Network Security
- Infrastructure Hardening
- Secure Deployments

Module 10: Responsible AI Governance & Compliance

Duration: 2 Weeks

Topics

- AI Governance Frameworks
- AI Lifecycle Governance
- AI Documentation
- Model Cards
- Datasheets for Datasets
- AI Audit Trails
- Risk Registers
- Compliance Overview
- Responsible AI Policies
- AI Incident Management

Module 11: AI Monitoring & Incident Response

Duration: 2 Weeks

Topics

- Production AI Monitoring
- Drift Detection
- Performance Monitoring
- Security Monitoring

- Log Analysis
- AI Incident Response
- Root Cause Analysis
- Continuous Improvement
- Threat Intelligence (Introduction)